O-S-377b
April 20, 1966

SUPERSEDING
Int. Fed. Spec. O-S-00377a (GSA-FSS)
March 11, 1964 and
Fed. Spec. O-S-377
February 3, 1954

FEDERAL SPECIFICATION

SODIUM PYROPHOSPHATE, ANHYDROUS, TECHNICAL (TETRASODIUM-PYROPHOSPHATE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sodium pyrophosphate (tetrasodium-pyrophosphate) for use in various washing, cleaning, and scouring processes with soap and other detergents where a mildly alkaline material, with appreciable water softening properties, is desired.

1.2 Classification.

1.2.1 *Grade.* The sodium pyrophosphate (tetrasodium-pyrophosphate) covered by this specification shall be furnished in one grade only.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-S-366—Sieves, Standard For Testing Purposes.
PPP-D-723—Drums, Fiber.
PPP-D-729—Drums: Metal, 55-Gallon (For Shipment of Noncorrosive Material).
PPP-P-704—Pails: Shipping, Steel (1 Through 12 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the

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procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Sodium pyrophosphate (tetrasodium-pyrophosphate) shall be a white uniform product in either granular or powdered form. It shall be clean and free from defects which may affect its suitability for various washing, cleaning, and scouring processes with soap or other detergents where a mildly alkaline material with water softening properties is required. The material shall conform to the chemical and physical requirements shown in table I.

TABLE I. *Chemical and physical requirements*

Requirement	Minimum	Maximum	Test method
Matter insoluble in water, percent by weight		0.2	4.4.1
Loss on ignition, percent	—	0.5	4.4.2
Assay (percent as $Na_4P_2O_7$) ...	98.0	—	4.4.3
Particle size passing No. 16 U. S. standard sieve, percent (RR-S-366)	99	—	4.4.4
Fe content	—	70 p.p.m.	4.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 A lot shall consist of the sodium pyrophosphate offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot. From each lot offered for delivery, two containers shall be taken at random by the Government inspector and opened. A sample of approximately 0.5 pound shall be taken from each selected container and placed in a clean, dry, airtight container, which shall be nearly filled. The inspector shall seal, mark, and accurately weigh each sample, record its weight, and date of weighing on the sample container and forward to a Government laboratory for test purposes. Samples shall be kept cool until tested.

4.3 Inspection.

4.3.1 *Sampling for inspection of containers (at the contractor's plant).* A random sample of filled containers shall be taken by the Government inspector in accordance with MIL-STD-105 at inspection level I and an acceptable quality level (AQL) of 2.5 percent defective to verify compliance with this specification regarding fill, closure, marking, and other requirements not involving tests.

4.3.2 *Inspection of containers (at the contractor's plant).* Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or under required fill, shall be rejected, and if the numer of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed or repaired all nonconforming containers.

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4.3.3 *Inspection of preparation for delivery requirements.* An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with table II. For examination of the contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE II. *Classification of preparation for delivery defects*

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as, incomplete closure of container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of container.

4.4 Test procedures.

4.4.1 *Matter insoluble in water.* Prepare a tared fitted Gooch or Gooch-type crucible in the following manner: The Gooch crucible should be uniformly lined with finely divided asbestos wadding, previously contained in distilled water, so as to cover all holes in the crucible. Dry the crucible in an oven to 110°C. Cool and weigh. Weigh 100 grams (g.) of the sample to the nearest 0.5 g. and transfer to a 1-liter beaker. Dissolve by stirring with 900 milliliters (ml.) of distilled water at 80° to 90°C. Filter by suction, using the prepared asbestos felted Gooch crucible. Wash the beaker and residue free from alkali with hot distilled water, and dry the crucible to constant weight in an oven at 110°C. Cool in a desiccator and weigh. Calculate the percentage of matter insoluble in water as follows:

$$\text{Percent matter insoluble in water} = \frac{\text{Weight of residue in grams}}{\text{Weight of sample}} \times 100$$

4.4.2 *Loss on ignition.* Weigh 2 g. of the sample into a porcelain crucible which has previously been ignited to constant weight. Heat in a muffle furnace at 700°C. for 30 minutes, cool in a desiccator and weigh.

Calculate the loss on ignition as follows:

$$\text{Loss on ignition, percent} = \frac{\text{Loss in weight}}{\text{Weight of sample}} \times 100$$

4.4.3 *Assay.* Assay shall be determined by method 1502.1 of Fed. Test Method Std. No. 536, omitting the alcohol extraction step.

4.4.4 *Fineness.* Fineness shall be determined by method 210.1 of Fed. Test Method Std. No. 536, using a No. 16 sieve.

4.4.5 *Iron content.*

4.4.5.1 *Standard iron solution.* Dissolve about 2.0 g. of ferric chloride ($FeCl_3 \cdot 6H_2O$) in a liter of distilled water containing 10 ml. of hydrochloric acid (sp. gr. 1.18). Determine the exact ferric iron content of this solution by a standard method. From a burette, run the calculated volume of this solution into a liter volumetric flask, to yield a concentration of 0.002 milligram (mg.) of iron per ml. Dilute to the mark. One ml. of the solution is equivalent to 20 p.p.m. of iron in a 0.1 g. sample.

4.4.5.2 *Iron content.* Weigh 1.00 g. of the sample in a 250 ml. beaker, add 15 ml. of nitric acid (sp. gr. 1.42), and 5 ml. of sulphuric acid (sp. gr. 1.84). Heat in a hood until dense fumes of sulphur trioxide are evolved and long enough to destroy any organic matter that might be present. Cool, add 30 ml. of water, 10 ml. of hydrochloric acid (sp. gr. 1.17) and 0.3 ml. of nitric acid (sp. gr. 1.42); boil 10 minutes, cool to room temperature, and transfer to a 100 ml.

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volumetric flask. (In case the iron content is low, 10 to 20 p.p.m., use the whole sample, dilute to 50 ml. with acetone, for the determination.) Dilute to volume, shake well, and transfer a suitable aliquot to a 50 ml. Nessler tube. Add 25 ml. of acetone and 5 ml. of hydrochloric acid (sp. gr. 1.17), mix well, using a clean stopper and allow to stand 10 minutes. Add 10 ml. of a 20 percent ammonium thiocyanate solution and after mixing, compare with freshly prepared standards, made by treating aliquots of the standard iron solution (see 4.4.5.1) in the manner described above and finally diluting to the same volume as the sample.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of levels of packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packing.** Packing shall be levels A, B, or C as specified (6.1).

5.1.1 *Levels A and B.* Unless otherwise specified, the product shall be packed in 50-, 100-, or 300-pound quantities as specified (6.1). The 50- and 100-pound quantities shall be packed in pails and drums conforming to PPP-P-704, type II; and PPP-D-723, type II, grade A. The 300-pound quantity shall be packed in a drum conforming to PPP-D-729, type III. Pails conforming to PPP-P-704 shall have an interior coating that will not have a deleterious effect on the pail or the product packed. Closures shall be as specified for the drum type in the applicable drum specification.

5.1.2 *Level C.* The product shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with rules and regulations applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Levels of packing required (see 5.1).

6.2 Sodium pyrophosphate should be purchased by net weight.

Review interest:

Army—MD, EL, MU

Air Force—68

Preparing activity:

GSA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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